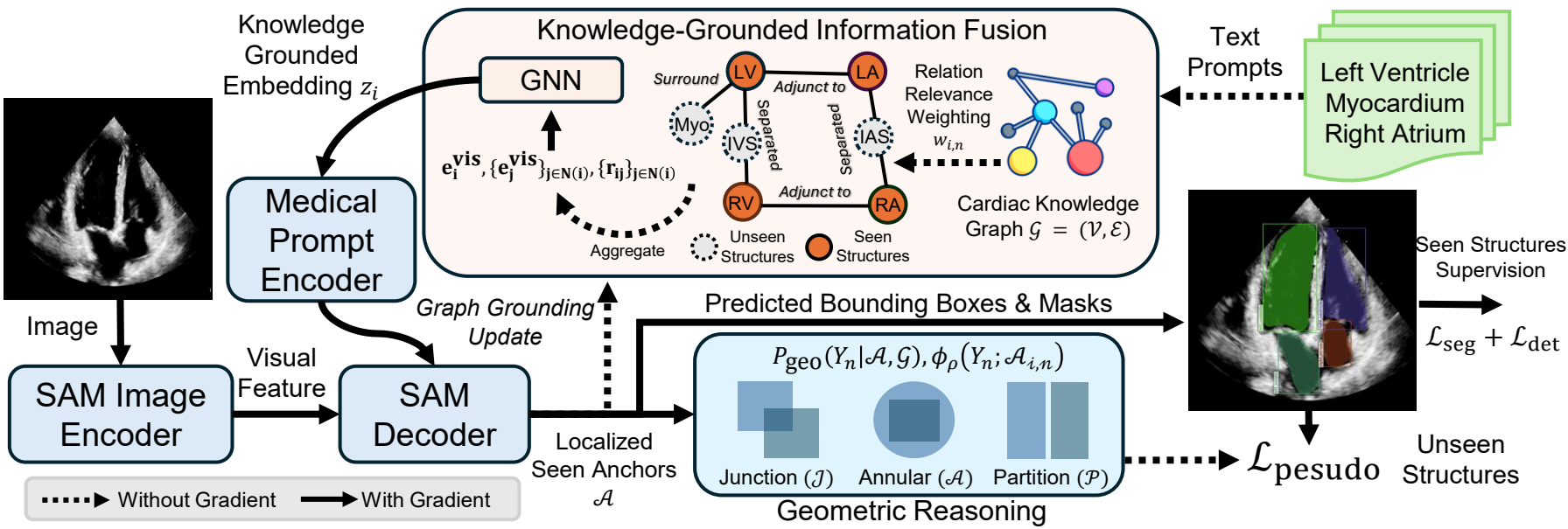

Knowledge Grounded Embedding z_i
Knowledge-Grounded Information Fusion
GNN
Surround
LV
Adjunct to
LA
Myo
IVS
Separated
Separated
IAS
RV
Adjunct to
RA
Unseen Structures
Seen Structures
$\mathbf{e}_i^{\mathbf{vis}}, \{\mathbf{e}_j^{\mathbf{vis}}\}_{j\in N(i)}, \{\mathbf{r}_{ij}\}_{j\in N(i)}$
Aggregate
Relation Relevance Weighting $w_{i,n}$
Cardiac Knowledge Graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$
Text Prompts
Left Ventricle
Myocardium
Right Atrium
Medical Prompt Encoder
Image
SAM Image Encoder
Visual Feature
SAM Decoder
Graph Grounding Update
Predicted Bounding Boxes & Masks
Seen Structures Supervision
$\mathcal{L}_{\text{seg}} + \mathcal{L}_{\text{det}}$
Localized Seen Anchors $\mathcal{A}$
$P_{\text{geo}}(Y_n|\mathcal{A}, \mathcal{G}), \phi_\rho(Y_n; \mathcal{A}_{i,n})$
Junction ($\mathcal{J}$)
Annular ($\mathcal{A}$)
Partition ($\mathcal{P}$)
Geometric Reasoning
$\mathcal{L}_{\text{pesudo}}$
Unseen Structures
Without Gradient
With Gradient